

lications) as a primary source, which provides insights about existing knowledge along with gaps and unanswered questions². Formally, let $\mathcal{L}$ be the literature, and $\mathbf{o}$ be the ideas that consist of the problem $\mathbf{p}$, method $\mathbf{m}$, and experiment design $\mathbf{d}$, as follows: $\mathbf{o} = [\mathbf{p}, \mathbf{m}, \mathbf{d}]$ where each item consists of a sequence of tokens. Then, the idea generation model f can be represented as follows: $\mathbf{o} = f(\mathcal{L})$, which is further decomposed into three submodular steps: $\mathbf{p} = f(\mathcal{L})$ for identifying problems, $\mathbf{m} = f(\mathbf{p}, \mathcal{L})$ for developing methods, and $\mathbf{d} = f(\mathbf{p}, \mathbf{m}, \mathcal{L})$ for designing experiments. We operationalize f with LLMs, leveraging their capability to understand and generate academic text.

Large Language Models Before describing the LLM in the context of our problem setup, let us first provide its general definition, which takes an input sequence of tokens $\mathbf{x}$ and generates an output sequence of tokens $\mathbf{y}$, as follows: $\mathbf{y} = \text{LLM}_\theta(\mathcal{T}(\mathbf{x}))$. Here, the model parameters θ are typically fixed after training, due to the high costs of further fine-tuning. In addition, the prompt template $\mathcal{T}$ serves as a structured format that outlines the context (including the task descriptions and instructions) to direct the model in generating the desired outputs.

3.2 Knowledge-Augmented LLMs for Research Idea Generation

We now turn to our primary focus of automatically generating research ideas with LLMs. Recall that we aim to produce a complete idea consisting of the problem, method, and experiment design ($\mathbf{o} = [\mathbf{p}, \mathbf{m}, \mathbf{d}]$), while using the existing literature $\mathcal{L}$ as a primary source of information. We operationalize this with LLMs by instantiating the aforementioned research idea generation function f with LLM coupled with the task-specific template. Formally, $\mathbf{p} = \text{LLM}(\mathcal{T}_p(\mathcal{L}))$ indicates the problem identification step, followed by $\mathbf{m} = \text{LLM}(\mathcal{T}_m(\mathbf{p}, \mathcal{L}))$ for method development and $\mathbf{d} = \text{LLM}(\mathcal{T}_e(\mathbf{p}, \mathbf{m}, \mathcal{L}))$ for experiment design, which constitutes the full idea: $\mathbf{o} = [\mathbf{p}, \mathbf{m}, \mathbf{d}]$.

Following this general formulation, the important question to answer is how the body of scientific literature is leveraged for actually generating research ideas with LLMs. Here, we outline three key desiderata that contribute to the success of human researchers ideating novel research ideas: a broad

and deep understanding of related work, an encyclopedic perspective on the interconnectedness of concepts within and across scientific domains, and a community of peers who help iteratively improve ideas through constructive critiques. We describe our operationalization of these three desiderata using the prior literature and LLMs in what follows.

Citation Graph-based Literature Survey Due to the constraints on their input lengths and their reasoning abilities, particularly over very long contexts (Liu et al., 2023a), it is not possible to incorporate all the existing publications from the literature $\mathcal{L}$ into the LLM input. Instead, we need to find a meaningful subset relevant to the problem at hand. To achieve this, we mirror the process followed by human researchers, who expand their knowledge of a paper by perusing other papers that either cite or are cited by it. Concretely, for the LLM, we initiate its literature review process by providing a core paper l_0 from $\mathcal{L}$ and then selectively incorporating subsequent papers $\{l_1, \dots, l_n\}$ that are directly connected based on a citation graph. This procedure makes the LLM input for idea generation more manageable and coherent. In addition, we operationalize the selection process of the core paper and its relevant citations with two design choices: 1) the core paper is selected based on its citation count (e.g., exceeding 100 over 3 months) typically indicating high impact; 2) its relevant papers (which may be potentially numerous) are further narrow-downed based on their similarities of abstracts with the core paper, ensuring a more focused and relevant set of related work.

However, despite the simplicity and intuitiveness of this idea generation approach, there exists one major limitation. This approach relies exclusively on a set of given papers (the core paper and its citations); however, since scientific knowledge is not confined to specific studies but rather accumulates across a wide range of publications (across various fields), we should ideally harness this extensive, interconnected, and relevant scientific knowledge in our method for research idea generation.

Entity-Centric Knowledge Augmentation In order to model an encyclopedic view of interconnected concepts, we must effectively design a framework to extract, store and effectively leverage the vast amount of knowledge in scientific literature $\mathcal{L}$. In this work, we view entities as the atomic units of knowledge, which allows for ease of representation and accumulation over papers in a unified

²We focus on the existing literature-based idea generation by following the paradigm that a *new idea* is more often than not just a new combination of old elements (Young, 2003).

manner across different disciplines. For example, we can easily extract the term “database” whenever it appears in any paper, using existing off-the-shelf entity linking methods and then aggregate their linked occurrences into a knowledge store. Then, if the term “database” is prevalent within the realm of medical science but less so in hematology (which is a subdomain of medical science), the constructed knowledge store can capture the affinity between those two domains based on overlapping entities. This representational paradigm can then be used to suggest the term “database” when formulating the ideas about hematology. In other words, this approach enables providing novel and interdisciplinary insights by leveraging the interconnectedness of entities across various fields.

Formally, we design the knowledge store as a two-dimensional matrix $\mathcal{K} \in \mathcal{R}^{m \times m}$ where m is the total number of unique entities identified and $\mathcal{K}$ is implemented in a sparse format. This knowledge store is constructed by extracting entities over all the available scientific articles in literature $\mathcal{L}^3$, which not only counts the co-occurrences between entity pairs within individual papers but also quantifies the count for each entity. Our approach is versatile, thus, we can use any entity linker; in this paper we use one developed by Wu et al. (2020). This off-the-shelf system proves capable of extracting key scientific entities (which is demonstrated in Table 16) despite its lack of customized training for the scientific domain. Specifically, this linker tags and canonicalizes entities in a paper l from $\mathcal{L}$, formalized as follows: $\mathcal{E}_l = \text{EL}(l)$ where $\mathcal{E}_l$ denotes a multiset of entities (allowing for repetitions) appearing in l^4 . Upon extracting entities $\mathcal{E}$, to store them into the knowledge store $\mathcal{K}$, we consider all possible pairs of $\mathcal{E}$ represented as follows: $\{e_i, e_j\}_{(i,j) \in \mathcal{C}(|\mathcal{E}|, 2)}$ where $e \in \mathcal{E}$.

Given this knowledge store $\mathcal{K}$, our next goal is to enhance the previous vanilla research idea generation process implemented based on a group of interconnected papers, denoted as follows: $\mathbf{o} = \text{LLM}(\mathcal{T}(\{l_0, l_1, \dots, l_n\}))$. We do this by augmenting the LLM with the relevant entities from $\mathcal{K}$, which expand the context that LLMs consume with additional knowledge. Formally, let us define entities extracted from the group of interconnected papers, as follows: $\mathcal{E}_{\{l_0, \dots, l_n\}} = \bigcup_{i=0}^n \text{EL}(l_i)$. Then, the

probabilistic form of retrieving the top- k relevant external entities can be represented as follows:

$$\text{Ret}(\{l_0, \dots, l_n\}; \mathcal{K}) = \arg \max_{I \subset [m]: |I|=k} \prod P(e_i | \mathcal{E}_{\{l_0, \dots, l_n\}}), \quad (1)$$

where $[m] = \{1, \dots, m\}$ and $e_i \notin \mathcal{E}_{\{l_0, \dots, l_n\}}$. Also, for simplicity, by applying Bayes’ rule and assuming that entities are independent, the retrieval operation (Equation 1) can be approximated as follows:

$$\arg \max_{I \subset [m]: |I|=k} \prod_{e_j \in \mathcal{E}_{\{l_0, \dots, l_n\}}} (P(e_j | e_i)) \times P(e_i), \quad (2)$$

where $P(e_j | e_i)$ and $P(e_i)$ can be derived from values in the two-dimensional matrix $\mathcal{K}$, suitably normalized. We note that the formulation in Equation 2 is only one instance of operationalizing retrieval; this could be replaced with other retrieval strategies – for example, embedding-based retrieval (discussions and results are provided in Appendix B.3). Hereafter, the instantiation of research proposal generation augmented with relevant entity-centric knowledge is formalized as follows: $\mathbf{o} = \text{LLM}(\mathcal{T}(\{l_0, \dots, l_n\}, \text{Ret}(\{l_0, \dots, l_n\}; \mathcal{K})))^5$. We call this knowledge-augmented LLM-powered idea generation approach *ResearchAgent*, and provide the templates to instantiate it in Tables 6, 7, and 8.

Iterative Research Idea Refinements We note that attempting to write a full research idea in one go may not be an effective strategy. Humans write drafts that are continually improved based on multiple rounds of reviews and feedback. Therefore, we lastly model a community of peers for iterative idea improvement by introducing a set of LLM-powered reviewing agents (called *ReviewingAgents*), which provide the *ResearchAgent* with reviews and feedback according to various criteria for improvement.

Specifically, similar to our approach to instantiate *ResearchAgent* with an LLM (LLM) and template ($\mathcal{T}$), *ReviewingAgents* are instantiated similarly but with different templates (See Tables 9, 10, and 11). Then, with *ReviewingAgents*, each of the generated research ideas (problem, method, and experiment design) is separately evaluated according to its own specific five criteria⁶, which are provided in labels of Figure 2 and detailed in Table 12. Based on the reviews and feedback from *ReviewingAgents*, the *ResearchAgent* iteratively updates and refines its generation of research ideas.

³As extracting entities on all articles is computationally infeasible, we target papers appearing after May 01, 2023.

⁴Due to the extensive length of scientific publications, the target of entity extraction is restricted to titles and abstracts.

⁵There may be additional knowledge sources (beyond the existing literature and entities) for research idea generation, and we leave exploring them as future work.

⁶We select the top five criteria which we consider as the most important, and leave exploring others as future work.

Despite the proficiency of LLMs in the evaluation of machine-generated texts (Zheng et al., 2023; Fu et al., 2023), their judgments on research ideas may not be aligned with the judgments of real human researchers. However, there are no ground truth reference judgments available, and collecting them to align LLMs is expensive and often infeasible. Ideally, the judgments made by LLMs should be similar to the ones made by humans, and we aim to ensure this by automatically generating human preference-aligned evaluation criteria (used for automatic evaluations) with a few human annotations. Specifically, to obtain these human-aligned evaluation criteria, we first collect 10 pairs of research ideas and their associated scores for every evaluation criterion on a 5-point Likert scale, annotated by human researchers having at least 3 papers. After that, we prompt the LLM with these human-annotated pairs to induce detailed descriptions for evaluation criteria (Lin et al., 2024) (See Tables 13, 14, and 15). These criteria reflect the underlying human preferences⁷ and are used as evaluation criteria by the ReviewingAgents.

4 Experimental Setup

4.1 Data

The main source to generate research ideas is the scientific literature $\mathcal{L}$, which we obtain from the Semantic Scholar Academic Graph API⁸. From this, we select papers appearing after May 01, 2023, because LLMs that we use in our experiments are trained on data from the open web available before this point. This follows the procedure of existing literature-based hypothesis generation work (Qi et al., 2023). Then, we select high-impact papers (that have more than 20 citations) as core papers, mirroring human researchers’ tendency to leverage influential work, to ensure the high quality of generated ideas. The resulting data is still very large; thus, we further sample a subset of 300 papers as core papers to obtain a reasonably sized benchmark dataset. The average number of reference papers for each core paper is 87; the abstract of each paper has 2.17 entities on average. The distribution of disciplines for all papers is provided in Figure 7.

⁷We additionally ask five human annotators, who evaluate research ideas, to judge the quality of the induced criteria; two of them agree strongly, while the other three agree moderately.

⁸<https://www.semanticscholar.org/product/api>

4.2 Baselines and Our Model

As we target the novel task of research idea generation involving the generation of problems, methods, and experimental designs, there are no baselines for direct comparison. Thus, we mainly compare our full ResearchAgent model against its ablated variants, outlined as follows:

1. **Naive ResearchAgent** – which uses only a core paper to generate research ideas.
2. **ResearchAgent w/o Entity Retrieval** – which uses the core paper and its relevant references without considering entities.
3. **ResearchAgent** – which is our full model that uses the relevant references and entities along with the core paper, to augment LLMs.

In addition to this set of core baselines, we also compare our approach against existing hypothesis generation work from prior literature in Table 3.

4.3 Evaluation Setup

Given our formulation of idea generation (Sec 3.1), there are no ground-truth answers to measure the quality of the generated ideas. Yet, exhaustively listing pairs of core papers and reference research ideas is suboptimal, since there may exist a large number of valid research ideas for each core paper, and this process requires much time, effort and expertise on the part of human researchers. Thus, we use a combination of model-based automatic evaluation and manual human evaluation to validate different models on our experimental benchmark.

Model-based Evaluation Following the recent trends in using LLMs to judge the quality of output texts (especially in the setting of reference-free evaluations) (Zheng et al., 2023; Fu et al., 2023; Liu et al., 2023b), we use GPT-4 to judge the quality of research ideas. Note that each of the problem, method, and experiment design is evaluated with five different criteria (See labels of Figure 2 for criteria and see Table 12 for their detailed descriptions). We ask the LLM-based evaluation model to either rate the generated idea on a 5-point Likert scale for each criterion or perform pairwise comparisons between two ideas from different models. We provide the prompts for evaluations in Appendix A.

Human Evaluation Similar to model-based evaluations, we perform human evaluations that involve assigning a score for each criterion and conducting pairwise comparisons between two ideas. As the